TASK ONE:

```sql
DROP TABLE IF EXISTS creatureType;
CREATE TABLE creatureType(
 creatureTypeID INTEGER PRIMARY KEY AUTO_INCREMENT,
 creatureType VARCHAR(20),
 creatureTypeDesc TINYTEXT
) ENGINE=InnoDB;

describe creatureType;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
creatureTypeID	int	NO	PRI	NULL	auto_increment
creatureType	varchar(20)	YES		NULL	
creatureTypeDesc	tinytext	YES		NULL	

```sql
INSERT INTO creatureType (creatureType, creatureTypeDesc) VALUES ("Person", "a member of the flawed human race");
INSERT INTO creatureType (creatureType, creatureTypeDesc) VALUES ("Frog", "a carnivorous, short-bodied, tailless amphibian");
INSERT INTO creatureType (creatureType, creatureTypeDesc) VALUES ("Kaiju", "giant monster that attacks major cities, the milit
INSERT INTO creatureType (creatureType, creatureTypeDesc) VALUES ("Superhero", "possesses superpowers and fits the role of the
INSERT INTO creatureType (creatureType, creatureTypeDesc) VALUES ("Villain", "a cruelly malicious person who is involved in or
INSERT INTO creatureType (creatureType, creatureTypeDesc) VALUES ("Sidekick", "a close companion or colleague who is generally

select * from creatureType;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

creatureTypeID	creatureType	creatureTypeDesc
1	Person	a member of the flawed human race
2	Frog	a carnivorous, short-bodied, tailless amphibian
3	Kaiju	giant monster that attacks major cities, the milit...
4	Superhero	possesses superpowers and fits the role of the ...
5	Villain	a cruelly malicious person who is involved in or d...
6	Sidekick	a close companion or colleague who is generally...

TASK TWO:

```
1  DROP TABLE IF EXISTS creature;
2  CREATE TABLE creature (
3      creatureId          INTEGER      PRIMARY KEY,
4      creatureName        VARCHAR(30),
5      creatureTypeID      int,
6      townID              int,
7
8      CONSTRAINT town_townID FOREIGN KEY (townID) REFERENCES town (townID),
9      CONSTRAINT creatureType_creatureTypeID FOREIGN KEY (creatureTypeID) REFERENCES creatureType (creatureTypeID)
10 )ENGINE=InnoDB;
11
12 describe creature
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
creatureId	int	NO	PRI	NULL	
creatureName	varchar(30)	YES		NULL	
creatureTypeID	int	YES	MUL	NULL	
townID	int	YES	MUL	NULL	

```
1      INSERT INTO creature VALUES (1,  'Bannon',1,1);
2      INSERT INTO creature VALUES (2,  'Myers',1,2);
3      INSERT INTO creature VALUES (3,  'Neff',1,3);
4      INSERT INTO creature VALUES (4,  'Neff',1,4);
5      INSERT INTO creature VALUES (5,  'Mieska',1,5);
6      INSERT INTO creature VALUES (6,  'Carlis',1,1);
7      INSERT INTO creature VALUES (7,  'Kermit',2,6);
8      INSERT INTO creature VALUES (8,  'Godzilla',3,7);
9      INSERT INTO creature VALUES (9,  'Thor',4,8);
10     INSERT INTO creature VALUES (10, 'The Tick',4,9);
11     INSERT INTO creature VALUES (11, 'Chairface Chippendale',5,9);
12     INSERT INTO creature VALUES (12, 'Arthur',6, 9);
13
14     select * from creature;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

creatureId	creatureName	creatureTypeID	townID
1	Bannon	1	1
2	Myers	1	2
3	Neff	1	3
4	Neff	1	4
5	Mieska	1	5
6	Carlis	1	1
7	Kermit	2	6
8	Godzilla	3	7
9	Thor	4	8
10	The Tick	4	9
11	Chairface Chi...	5	9
12	Arthur	6	9
NULL	NULL	NULL	NULL

```sql
select c.creatureName, ct.creatureType, ct.creatureTypeDesc
from creature as c
join creaturetype as ct
on c.creatureTypeID = ct.creatureTypeID;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

creatureName	creatureType	creatureTypeDesc
Bannon	Person	a member of the flawed human race
Myers	Person	a member of the flawed human race
Neff	Person	a member of the flawed human race
Neff	Person	a member of the flawed human race
Mieska	Person	a member of the flawed human race
Carlis	Person	a member of the flawed human race
Kermit	Frog	a carnivorous, short-bodied, tailless amphibian
Godzilla	Kaiju	giant monster that attacks major cities, the milit...
Thor	Superhero	possesses superpowers and fits the role of the ...
The Tick	Superhero	possesses superpowers and fits the role of the ...
Chairface Chi...	Villain	a cruelly malicious person who is involved in or d...
Arthur	Sidekick	a close companion or colleague who is generally...

TASK THREE:

```
1  DROP TABLE IF EXISTS weapon;
2  create table weapon (
3      weaponID        integer primary key auto_increment,
4      weaponName      varChar(30),
5      weapontType     varChar(20),
6      damageType      varChar(20),
7      weaponRange     int
8  )ENGINE=InnoDB;
9
10 describe weapon;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
weaponID	int	NO	PRI	NULL	auto_increment
weaponName	varchar(30)	YES		NULL	
weapontType	varchar(20)	YES		NULL	
damageType	varchar(20)	YES		NULL	
weaponRange	int	YES		NULL	

```
1  insert into weapon (weaponName, weapontType, damageType, weaponRange) values ("club","melee","bludgeoning",2);
2  insert into weapon (weaponName, weapontType, damageType, weaponRange) values ("dagger","melee","piercing",2);
3  insert into weapon (weaponName, weapontType, damageType, weaponRange) values ("javelin","ranged","piercing",30);
4  insert into weapon (weaponName, weapontType, damageType, weaponRange) values ("great sword","melee","bludgeoning",10);
5  insert into weapon (weaponName, weapontType, damageType, weaponRange) values ("cross bow","ranged","piercing",40);
6  insert into weapon (weaponName, weapontType, damageType, weaponRange) values ("long bow","ranged","piercing",60);
7  insert into weapon (weaponName, weapontType, damageType, weaponRange) values ("mind control","magic","psychic",0);
8  insert into weapon (weaponName, weapontType, damageType, weaponRange) values ("read thoughts","magic","psychic",0);
9  insert into weapon (weaponName, weapontType, damageType, weaponRange) values ("eldritch blast","magic","bludgeoning",30);
10
11 select * from weapon;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

weaponID	weaponName	weapontType	damageType	weaponRange
1	club	melee	bludgeoning	2
2	dagger	melee	piercing	2
3	javelin	ranged	piercing	30
4	great sword	melee	bludgeoning	10
5	cross bow	ranged	piercing	40
6	long bow	ranged	piercing	60
7	mind control	magic	psychic	0
8	read thoughts	magic	psychic	0
9	eldritch blast	magic	bludgeoning	30
NULL	NULL	NULL	NULL	NULL

1. What type of relationship exists between creature and weapon? Looking at the description above you are only given this simple description: “a creature can be in possession of one or more weapons at any given time”. You will need to establish that the relationship is 1:M, M:N or 1:1. There is a correct answer to this. Let’s chat if you are unsure.

- The creature and weapon table have a many to many relationship (M:N)

2. Express the relationship using simple sentences. (include these sentences as part of your submission document)

- One creature can have multiple weapons, and one weapon can be used by multiple creatures.

```
1  create table creature_weapon (
2      creatureID int,
3      weaponID int,
4      primary key (creatureID, weaponID),
5      foreign key (creatureID) references creature(creatureID) on delete cascade,
6      foreign key (weaponID) references weapon(weaponID) on delete cascade
7  );
8
9  describe creature_weapon;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
creatureID	int	NO	PRI	NULL	
weaponID	int	NO	PRI	NULL	

```
1  select * from creature_weapon;
```

Result Grid | Filter Rows:

creatureID	weaponID
1	1
NULL	NULL

```
1  select * from creature as c
2  join creature_weapon as cw on c.creatureID = cw.creatureID
3  join weapon as w on cw.weaponID = w.weaponID;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

creatureId	creatureName	creatureTypeID	townID	creatureID	weaponID	weaponID	weaponName	weapontType	damageType	weaponRange
1	Bannon	1	1	1	1	1	club	melee	bludgeoning	2